B.Sc. in Computer Science and Engineering Thesis

Script Matters: Measuring Latin-Script Banglish Robustness in Bangla LLMs

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CANDIDATES’ DECLARATION

This is to certify that the work presented in this thesis, titled, “Script Matters: Measuring Latin-Script Banglish Robustness in Bangla LLMs”, is the outcome of the investigation and research carried out by us under the supervision of Dr. Sadia Sharmin.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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ABSTRACT

Bangla users do not always meet language models through the script assumed by standard benchmarks. The same Bengali question may be typed in native Bengali script, in English, or in Latin-script Banglish. Recent Bengali benchmarks increasingly cover knowledge, education, culture, inference, and social interaction, but they rarely isolate this script choice while holding the underlying task and answer fixed. This thesis studies whether orthography itself changes large language model behavior. We construct controlled Bangla, reviewed Banglish, and English variants of curriculum-style QA and math items drawn from BEnQA and BanglaMATH, and evaluate models under a paired item-level protocol.

On a 200-item validation slice, compact Qwen models show a consistent reviewed-Banglish deficit. Qwen2.5-3B scores 54/200 in Bangla but 41/200 in reviewed Banglish, Qwen2.5-7B 8-bit scores 65/200 versus 47/200, and Qwen3-4B scores 80/200 versus 49/200. The gap is not explained by token count alone, since reviewed Banglish is token-cheaper than native Bangla for the audited Qwen tokenizers. A frozen v5 review of high-impact Banglish rows changes scores by at most two items, improving benchmark quality without removing the script-conditioned weakness.

Hosted-model and scale checks clarify the boundary. GPT-5.5 low nearly closes the validation-200 gap under secondary scoring, while Claude Sonnet 4.6, DeepSeek V4 Flash, and Groq-hosted Llama 3.3 70B preserve reviewed-Banglish deficits under the same protocol. A 1,000-row human review audit extends the BEnQA evidence base; 974 accepted or edited rows form a gold extension, giving 1,174 usable human-reviewed paired items when combined with the 200-item core. On this extension, Qwen2.5-3B, Groq-hosted Llama 3.3 70B, and DeepSeek V4 Flash all reproduce the English greater-than Bangla greater-than reviewed Banglish ordering. Mitigation remains model-dependent: self-normalization helps one Qwen model and hurts another, and generated alternate-script views require preservation gates before they can support a deployable routing claim. Overall, the thesis shows that Latin-script Banglish is an undermeasured orthographic robustness challenge for Bangla LLM use. It is a real access path for Bangla users, and its reliability must be measured directly rather than assumed from native-script Bangla or English performance.

Chapter 1

Introduction

Bangla is normally written in Bengali script. In everyday digital use, however, many speakers also write Bengali-language content with Latin characters. This Latin-script form is often called Banglish. It appears in messaging, search queries, comments, reviews, and informal learning contexts, often because Latin keyboards are convenient or because the surrounding conversation is already code-mixed. For a Bangla-facing language model, this means the same user intent can arrive through more than one written form.

That small interface detail can matter. A student may ask the same science question in Bangla script, in English, or in Banglish. A model may answer one version and fail another, even though the gold answer has not changed. If we only evaluate native-script Bangla and English, this access gap can remain hidden.

This thesis asks a simple question:

If the task and gold answer are held fixed, does changing the script to Latin-script Banglish change model reliability?

This is not the same as asking whether Bangla is hard for LLMs. Existing Bengali benchmarks already show that Bangla evaluation is important. BEnQA evaluates Bengali and English science questions from Bangladeshi educational material [1]. BanglaMATH evaluates Bengali math word problems [2]. Other work expands Bengali evaluation across reading comprehension, multitask knowledge, cultural understanding, inference, safety, and social interaction [3–8]. These benchmarks establish that Bengali LLM evaluation is active and necessary.

The missing piece is narrower. Most benchmarks do not ask whether the same Bangla content remains answerable when written in Banglish. This matters because Banglish is not an academic curiosity. BanglaTLit documents romanized Bangla as a large back-transliteration problem with substantial spelling variation [9]. BanglishRev, BAN-TH, BnSentMix, MixSarc, and other datasets show that Bangla-English, code-mixed, and transliterated Bangla appear in reviews, social media, sentiment, hate-speech, humor, and sarcasm settings [10–13]. Romanized Indic language identification is itself treated as an infrastructure problem [14]. If models fail on Banglish, then a large class of users can receive lower-quality answers even when their request is semantically equivalent to a request the model can answer in another script.

We therefore frame Banglish as an **orthographic access problem**. The central claim is not that Banglish is always harder for every model or every item. The claim is that script choice is a measurable robustness variable: it can change accuracy, strict answer compliance, recoverability, and cost under controlled conditions.

The study is built around three design principles.

First, evaluation must be paired. We preserve the same item id and gold answer across Bangla, reviewed Banglish, and English variants. Aggregate accuracy alone is not enough; we need to know whether the same item becomes wrong under one script and correct under another.

Second, the Banglish variant must be audited. A rule-based romanizer is useful for controlled construction, but it can introduce unnatural or ambiguous forms. We therefore freeze a reviewed-v5 validation slice after targeted human review of high-impact Banglish rows. We also keep a strict policy for flagged source quality issues and report a stricter 197-row sensitivity view.

Third, stronger models should be treated as a boundary test, not as a shortcut. If GPT-5.5 nearly closes the gap, that does not invalidate the robustness finding; it tells us where the current boundary is. Conversely, if Claude, DeepSeek, or Groq-hosted Llama still show Banglish deficits, then model or provider prestige alone is not a reliable guarantee of Banglish robustness.

In the following chapters, we first describe the paired benchmark construction and scoring protocol. We then present the main validation-200 result, test whether it survives review and tokenization controls, examine recoverable failures, and finally study hosted frontier models, the larger BEnQA extension, natural code-mixed text, and mitigation attempts.

1.1 Contributions

This thesis makes five contributions.

1. **A paired Bangla/Banglish/English evaluation protocol.** We evaluate the same curriculum-style QA and math items in three script views while preserving the item id, answer type, and gold answer.

2. **A reviewed gold-core validation set.** The main validation-200 v5 slice contains 144 BEnQA and 56 BanglaMATH items. Its reviewed Banglish field is frozen after a targeted

review pass, and all main claims use the same all-200 denominator.

3. **Evidence that Banglish failure is not just cleanup or token length.** Human review changes scores only slightly. Tokenization audits show that reviewed Banglish is shorter than native Bangla for the audited Qwen tokenizers, and recoverable Banglish misses are not the longest Banglish prompts.

4. **A frontier and scale boundary.** A five-model API panel shows that GPT-5.5 nearly closes the gap on the 200-item core, but Gemini, Claude, DeepSeek, and Groq show different residual failure profiles. A 1,000-row human review audit creates a 974-row accepted/edited BEnQA gold extension, with full triad runs for Qwen2.5-3B, Gemini 3.5 Flash, GPT-5.5, Claude Sonnet 4.6, DeepSeek V4 Flash, and Groq-hosted Llama 3.3 70B. At this scale, every model — including GPT-5.5 — shows a statistically significant reviewed-Banglish deficit.

5. **Negative mitigation evidence.** Self-normalization, generated Bengali views, and generated English views do not produce a general deployable solution. Preservation gates are necessary, and current cheap generated views are too unstable for a held-out routing claim. A training-side LoRA adapter, trained on romanized supervision disjoint from both evaluation sets, narrows the gap only slightly and not significantly, indicating the weakness is deeper than light adaptation.

Chapter 2

Related Work

2.1 Bengali evaluation

BEnQA introduced a Bengali-English question answering benchmark based on science questions from Bangladeshi education [1]. BanglaMATH introduced a Bengali math word-problem benchmark and showed language bias in mathematical reasoning [2]. MGSM provides multilingual GSM8K-style arithmetic items and supports broader multilingual reasoning evaluation [15, 16].

Recent Bengali benchmarks cover many other directions: open-domain QA [3], multitask knowledge [4], linguistic and cultural knowledge [5], textbook QA [6], inference [7], social and cultural alignment [8], multimodal cultural understanding [17], long-form speech [18], safety [19], and biomedical QA [20]. This growing landscape is important for positioning our work. We do not claim that Bengali is unevaluated. The gap is more specific: controlled downstream Bangla/Banglish/English script-equivalence evaluation remains undermeasured.

2.1.1 Romanized Bangla and code-mixed text

BanglaTLit shows that romanized Bangla is widespread and spelling-variable, requiring dedicated back-transliteration resources [9]. BanglishRev documents Bangla-English and code-mixed product reviews [10]. BAN-TH focuses on transliterated Bangla hate speech [11], BnSentMix on Bengali-English code-mixed sentiment [12], and MixSarc on implicit meaning in Bangla-English code-mixed text [13]. These resources show that Banglish and related code-mixed forms are practical user-facing formats.

Most of these datasets are classification corpora. They are useful prevalence evidence, but they do not directly measure whether a QA or math item with the same answer becomes harder when the script changes. That paired downstream question is the core of Script Matters.

2.1.2 Script robustness and transliteration

Prior work on Bangla transliteration perturbations shows that replacing Bangla words or sentences with transliterated forms can expose model vulnerabilities [21]. Related transliteration resources also motivate normalization as a possible bridge for Romanized Bengali inputs [22]. Recent work on romanized scripts in Indian-language triage and Romanized Nepali LLM evaluation supports the broader claim that romanized South Asian language inputs deserve direct evaluation [23, 24].

Our work differs in three ways. We use full task-equivalent variants rather than local perturbations. We preserve the item and gold answer across Bangla, Banglish, and English. And we analyze failure recoverability, tokenization, review sensitivity, frontier-model behavior, and mitigation attempts under one protocol. Table 2.1 summarizes the comparison with the three nearest neighbors.

Table 2.1: Comparison with the nearest prior work on romanized South Asian script robustness. "Partial" marks design elements present in a weaker form (word/sentence-level perturbation rather than full task-equivalent variants; automatic rather than human-reviewed transliteration).

Work	Paired items	Human-reviewed romanization	Downstream QA/math	Mitigation study	Scale extension
Transliteration perturbations [21]	Partial	No	No	No	No
Script-gap triage [23]	Yes	No	No	No	No
Romanized Nepali [24]	Partial	No	No	Yes	No
This thesis	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

The perturbation study replaces words or salient tokens inside existing classification datasets, so its paired contrast is local rather than item-equivalent, and it does not evaluate answer-preserving QA or math tasks. The script-gap triage study pairs native and romanized versions of real clinical queries but evaluates a single classification task without human review of the romanized text or any mitigation analysis. The Romanized Nepali study fine-tunes 8B models on transliterated instruction data — a training-side adaptation similar in spirit to our mitigation chapter — but evaluates generation fluency metrics rather than gold-answer task accuracy. None of the three combines paired gold-answer evaluation, human-reviewed romanization, multi-model downstream measurement, mitigation probes, and a human-reviewed scale extension under one protocol.

2.1.3 Tokenization and latent language

Tokenizer design can create unequal costs and context budgets across languages [25]. At the same time, multilingual LLMs can show latent English or romanized-language behavior [26–28]. These findings motivate our tokenization and cross-script oracle analyses. However, they do not imply that explicit Banglish input will be robust. Our results show the opposite for many models: Banglish can be token-cheaper than native Bangla and still less accurate.

Chapter 3

Benchmark Construction and Evaluation Protocol

3.1 Benchmark Construction

Figure 3.1 summarizes the full protocol: two source tasks feed a rule-based romanizer, targeted human review freezes the validation-200 v5 core, a separately sampled and fully human-reviewed 974-row BEnQA extension provides scale, and both slices are evaluated with the same paired triad protocol.

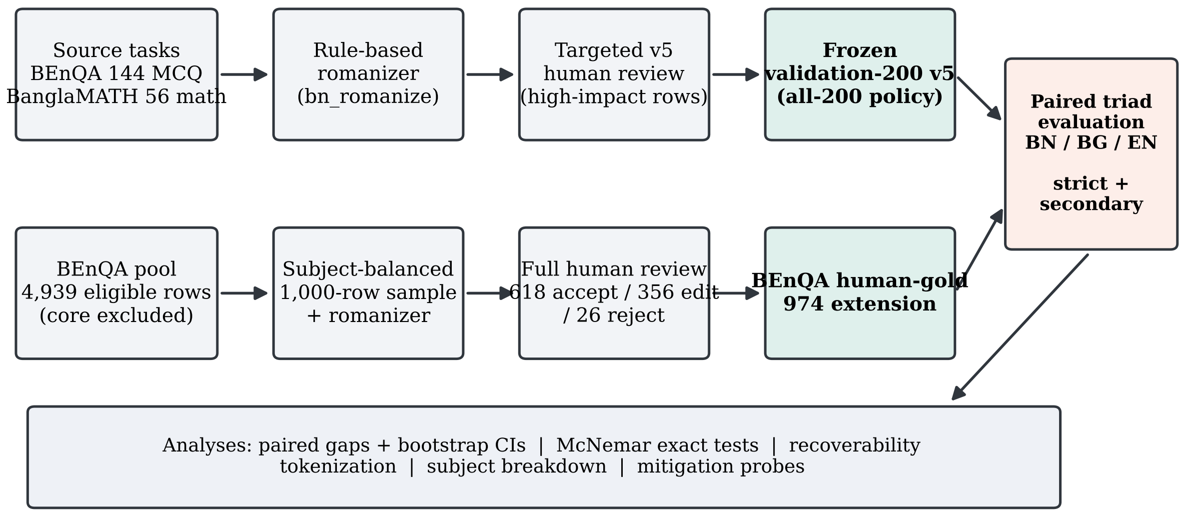

Figure 3.1: Benchmark construction and evaluation pipeline. The frozen validation-200 v5 core (top track) and the BEnQA human-gold 974 extension (bottom track) are evaluated under the same paired Bangla/Banglish/English protocol and shared analyses.

3.1.1 Source tasks

The benchmark construction has one goal: make script the main variable, not the question. To do this, the validation set combines two source tasks whose items can be represented in multiple script views while preserving the answer.

BEnQA. BEnQA contributes multiple-choice science questions. These items are well suited to paired script evaluation because the answer is a choice label and the same item can be represented in Bangla, Banglish, and English.

BanglaMATH. BanglaMATH contributes short-answer math word problems. These items stress numerical reasoning and answer normalization. They are harder to score strictly because correct answers may appear with units, Bengali digits, or intermediate reasoning.

The frozen validation-200 v5 slice contains 144 BEnQA items and 56 BanglaMATH items. The BEnQA subset is the cleanest source of script-gap evidence because choice-label scoring is less sensitive to answer formatting. BanglaMATH is kept as a stress-test slice because it exposes answer-compliance and numeric-unit issues that are important for deployment.

3.1.2 Script variants

Each item is represented in three main variants:

- **Bangla:** native Bengali script.
- **Reviewed Banglish:** Latin-script Banglish after the v5 review pass.
- **English:** English translation or English source variant.

All variants preserve the item id and gold answer. The prompt asks the model to return only the answer. For BEnQA, the expected output is one of A, B, C, or D. For BanglaMATH, the parser accepts short numeric or textual answers under strict and secondary modes. In other words, a script gap means the same item changes correctness when written differently; it does not mean that different questions were compared.

The English variant is useful but not identical in status to the Bangla-Banglish pair. English is a privileged comparison view. It helps identify whether an item is semantically answerable for a model, but the main orthographic claim is the Bangla-vs-reviewed-Banglish contrast.

3.1.3 Review and denominator policy

The v5 review pass targeted Banglish rows most likely to affect the thesis claim. After review, validation-200 v5 is frozen under an all-200 denominator policy. Three source-quality rows are also tracked under a strict-197 sensitivity view. The all-200 denominator remains primary because it avoids changing the evaluation set after seeing outcomes. The strict-197 view is a robustness check, not a replacement for the main result.

Review changed the main Banglish scores only slightly:

Table 3.1: Review sensitivity from v4 Banglish to reviewed-v5 Banglish.

Model	v4 Banglish	reviewed-v5 Banglish	Change
Qwen2.5-3B	39/200	41/200	+1.0 pts
Qwen2.5-7B 8-bit	48/200	47/200	-0.5 pts
Qwen3-4B	47/200	49/200	+1.0 pts

This is important. The review improves dataset quality but does not erase the script-conditioned weakness.

3.2 Evaluation Protocol

3.2.1 Models

The main open-model evaluation uses three compact Qwen instruction models:

- Qwen2.5-3B-Instruct
- Qwen2.5-7B-Instruct in 8-bit
- Qwen3-4B-Instruct

These models are small enough to run reproducibly on the available compute but strong enough to avoid a trivial "all models fail everything" result. They are therefore useful as thesis-facing baselines: failures are interpretable, but the experiments remain repeatable.

The frontier/API panel adds five hosted models:

- Gemini 3.5 Flash
- GPT-5.5 low
- Claude Sonnet 4.6

- DeepSeek V4 Flash, non-thinking mode
- Groq-hosted Llama 3.3 70B

The API rows use the same frozen validation-200 v5 prompt manifest and the same parser. They are not treated as a leaderboard. Their role is to test whether the Banglish gap disappears under stronger or differently trained systems.

3.2.2 Scoring

We report strict accuracy for all rows. Strict scoring measures whether the model produced the expected answer in a deployable form. For API rows we also report secondary parser/unit sensitivity. Secondary scoring gives credit for recoverable noncanonical answers, such as numeric-only answers with units or choice answers embedded in light markdown.

This separation is deliberate. Strict accuracy answers one deployment question: can the model follow the benchmark contract? Secondary accuracy answers a diagnostic question: did the model appear to know the semantic answer even if it violated the exact format?

3.2.3 Paired gaps

For each model and item, we compare correctness across script variants. The main gap is:

```
reviewed Banglish accuracy - Bangla accuracy
```

We also report:

```
reviewed Banglish accuracy - English accuracy
```

For the core Qwen rows and the human-reviewed BEnQA extension, we use paired bootstrap intervals. For the API validation panel, we also report exact paired sign-test style discordance summaries where available. Positive values mean Banglish is more accurate than the comparator. Negative values mean Banglish is less accurate.

Chapter 4

Main Results and Robustness Checks

4.1 Main Result: Reviewed Banglish Remains Harder

Table 4.1 gives the main frozen validation-200 v5 result.

Table 4.1: Main validation-200 v5 results

Model	Bangla	Reviewed Banglish	English	Banglish - Bangla	Banglish - English
Qwen2.5-3B	54/200	41/200	71/200	-6.5 pts, CI [-13.0, 0.0]	-15.0 pts, CI [-22.0, -7.5]
Qwen2.5-7B 8-bit	65/200	47/200	94/200	-9.0 pts, CI [-16.0, -2.0]	-23.5 pts, CI [-31.0, -16.0]
Qwen3-4B	80/200	49/200	88/200	-15.5 pts, CI [-22.0, -9.0]	-19.5 pts, CI [-27.0, -12.0]

The result is consistent across the three thesis-facing Qwen models. Reviewed Banglish is lower than native-script Bangla and English for every model. Qwen2.5-7B and Qwen3-4B have fully negative Banglish-minus-Bangla confidence intervals. Qwen2.5-3B has a negative point estimate but the all-200 interval touches zero. This is why the claim must be stated carefully: the strongest all-200 evidence comes from Qwen2.5-7B and Qwen3-4B, while Qwen2.5-3B remains supportive through point estimate, historical v3 results, and strict-197 sensitivity.

The English comparison is larger. For all three Qwen rows, reviewed Banglish is 15 to 23.5 points below English. This supports the idea that many failures are not simply item difficulty. The same item can become answerable when shown in a different script or language view.

Figure 4.1 shows the same pattern visually, together with the frontier/API panel from Chapter 6: reviewed Banglish is the lowest bar for every model except GPT-5.5, and the Banglish-minus-Bangla interval excludes zero for seven of the eight models.

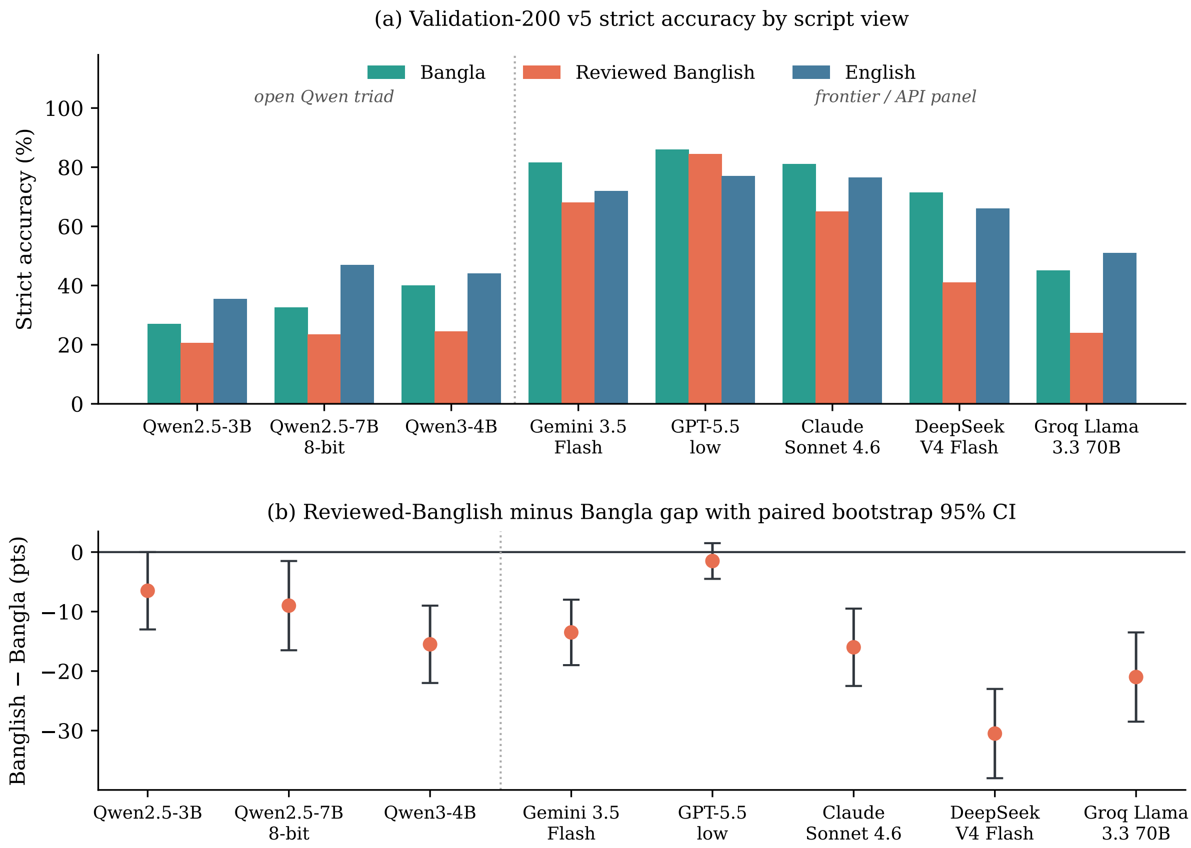

Figure 4.1: Validation-200 v5 strict accuracy by script view (top) and the reviewed-Banglish minus Bangla gap with paired bootstrap 95% confidence intervals (bottom) for the open Qwen triad and the five-model frontier/API panel.

4.2 Exact Paired Significance: McNemar Tests

The paired design is the textbook McNemar setting: each item produces a paired binary outcome under two prompt scripts. The exact McNemar test uses only the discordant pairs — items where exactly one script view is correct — so it is insensitive to the large mass of items that both views answer identically. We report the discordant counts (b = Bangla-only correct, c = Banglish-only correct), the exact two-sided p-value, the conditional odds ratio b/c with a Haldane-corrected 95% interval, and Holm-adjusted p-values across the three models.

Table 4.2: McNemar exact tests for reviewed Banglish versus Bangla on validation-200 v5. b counts Bangla-only-correct items; c counts Banglish-only-correct items.

Model	b	c	Odds ratio [95% CI]	Exact p	Holm p
Qwen2.5-3B	28	15	1.84 [0.99, 3.41]	0.066	0.066
Qwen2.5-7B 8-bit	37	19	1.92 [1.11, 3.32]	0.022	0.044
Qwen3-4B	39	8	4.65 [2.22, 9.75]	$<$0.0001	$<$0.0001

The exact tests sharpen the bootstrap story. Qwen3-4B and Qwen2.5-7B remain significant after

Holm correction. Qwen2.5-3B is marginal on the 200-item core ($p = 0.066$), which matches its CI-touches-zero bootstrap interval; on the 974-row BEnQA extension in Chapter 6, the same model's Bangla-versus-Banglish contrast becomes significant ($p = 0.024$), confirming that the marginal core result reflects sample size rather than absence of effect. The Banglish-versus-English contrasts are exact-test significant for all three models ($p \leq 0.0001$). Full tables for all panels, including the API panel and the extension, are in `reports/mcnemar_script_gaps.md`.

4.3 Per-Task Breakdown: BEnQA versus BanglaMATH

The two source tasks contribute differently to the aggregate result, so Table 4.3 splits the main result by task.

Table 4.3: Validation-200 v5 results split by source task. Gaps are reviewed Banglish minus Bangla with paired bootstrap 95% CIs.

Task	Model	Bangla	Reviewed Banglish	English	Banglish – Bangla
BEnQA (144)	Qwen2.5-3B	49/144	41/144	66/144	-5.6 pts, CI [-13.9, +2.8]
BEnQA (144)	Qwen2.5-7B 8-bit	60/144	47/144	86/144	-9.0 pts, CI [-18.8, +0.7]
BEnQA (144)	Qwen3-4B	76/144	47/144	82/144	-20.1 pts, CI [-28.5, -11.8]
BanglaMATH (56)	Qwen2.5-3B	5/56	0/56	5/56	-8.9 pts, CI [-17.9, -1.8]
BanglaMATH (56)	Qwen2.5-7B 8-bit	5/56	0/56	8/56	-8.9 pts, CI [-16.1, -1.8]
BanglaMATH (56)	Qwen3-4B	4/56	2/56	6/56	-3.6 pts, CI [-8.9, 0.0]

The BEnQA subset carries the cleanest evidence, as anticipated in Chapter 3: choice-label scoring is insensitive to answer formatting, and the Qwen3-4B BEnQA gap is the largest and most precisely estimated single contrast in the core (-20.1 points). The BanglaMATH subset operates near the strict-scoring floor — the compact Qwen models answer almost no Banglish math item in deployable form — so its gaps are smaller in absolute points but reflect a complete loss of usable accuracy under Banglish input. The two tasks therefore tell complementary stories: BEnQA measures a graded semantic gap, while BanglaMATH shows strict answer compliance collapsing under script change.

4.4 Review Sensitivity

A natural first objection is that the Latin-script Banglish field might simply be bad Banglish. The v5 review was designed to test that possibility. It targeted rows where review could matter most, then froze the final validation-200 slice. The result is clear: review changes scores only slightly and does not remove the gap.

This matters for interpretation. If Banglish cleanup had raised scores to match Bangla, the result would mostly be about romanizer artifacts. Instead, cleanup improves quality while preserving the central pattern. The study therefore does not need to claim that every romanization is perfect. It only needs to show that the observed gap is not removed by targeted review.

We also report what happens when the three flagged source-quality rows are excluded. Under this strict-197 policy, the reviewed Banglish-minus-Bangla intervals remain negative for all three thesis-facing Qwen rows:

Table 4.4: Strict-197 sensitivity check after excluding three flagged source-quality rows.

Model	Strict policy	Reviewed Banglish - Bangla
Qwen2.5-3B	197 rows	-7.1 pts, CI [-13.2, -1.0]
Qwen2.5-7B 8-bit	197 rows	-9.6 pts, CI [-16.8, -2.5]
Qwen3-4B	197 rows	-15.7 pts, CI [-22.3, -9.6]

The strict-197 view is not the primary denominator. It is a check that the main result does not depend on a few questionable rows.

4.5 Tokenization Sensitivity

Another plausible explanation is tokenization. If Banglish were much longer than Bangla, lower accuracy could be a context-budget or tokenization-efficiency problem. The data do not support that simple explanation.

For the audited Qwen tokenizers, reviewed Banglish is token-cheaper than native Bangla.

Table 4.5: Mean tokens per word under the audited Qwen tokenizers

Dataset	Bangla	Reviewed Banglish	English
BEnQA	4.02	2.49	1.95
BanglaMATH	4.63	2.11	1.41

Figure 4.2 makes the point visually: for every Qwen model, the reviewed-Banglish point sits to the left of the Bangla point (cheaper in tokens per word) and below it (less accurate). Token cost and accuracy are decoupled across scripts.

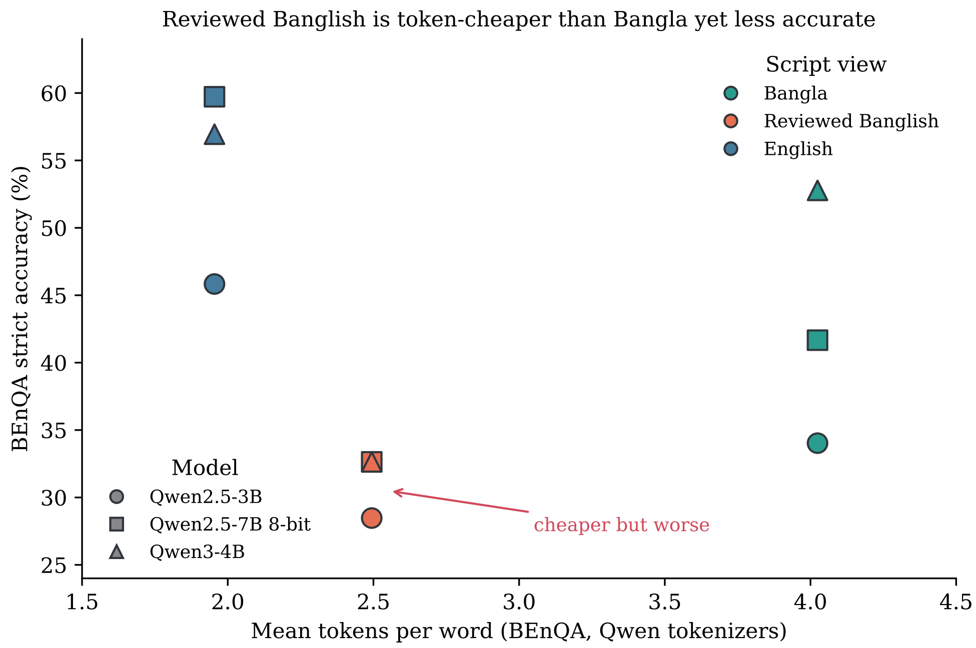

Figure 4.2: Token cost versus BEnQA strict accuracy for the Qwen triad on validation-200 v5. Reviewed Banglish costs fewer tokens per word than native Bangla under the audited Qwen tokenizers, yet is less accurate for every model.

The failure-pattern join also shows that recoverable Banglish misses are not the longest Banglish prompts. In BEnQA, recoverable reviewed-Banglish misses are shorter on average than other rows for all three thesis-facing Qwen models.

This does not prove tokenization is irrelevant. Tokenization can affect representation quality in ways not captured by length. But it rules out the simple explanation that Banglish fails because it uses more tokens. The failure is more consistent with script-conditioned lexical grounding, training distribution, spelling variation, and task interpretation.

4.6 Prompt and Decoding Sensitivity

A remaining objection is that the gap is an artifact of one prompt template or greedy decoding. We test this by rerunning Qwen2.5-3B on validation-200 v5 under two neutral alternate prompt templates — one terse, one exam-role-framed, neither containing any Banglish hint — and a temperature-0.7 decoding variant.

Table 4.6: Prompt and decoding sensitivity for Qwen2.5-3B on validation-200 v5. The Banglish-minus-Bangla gap stays negative across all conditions.

Condition	Bangla	Banglish	English	Banglish – Bangla
Baseline (greedy)	54/200	41/200	71/200	-6.5 pts
Neutral template B (terse)	51/200	45/200	71/200	-3.0 pts
Neutral template C (exam role)	53/200	45/200	74/200	-4.0 pts
Temperature 0.7	53/200	44/200	71/200	-4.5 pts

The Banglish-minus-Bangla gap stays negative (-3.0 to -6.5 points) under every template and the temperature change, and the Banglish-minus-English gap stays large (-13.0 to -15.0 points). The effect is therefore not produced by a particular instruction wording or by greedy decoding; it survives the same way the review, strict-197, and tokenization checks survive.

4.7 Spelling-Variation Robustness

The clearest stated limitation of the controlled benchmark is that one rule-based romanizer is not natural, spelling-variable Banglish. We test whether the script penalty depends on the specific spelling by generating, for a 100-item BEnQA subset, four additional seeded phonetic respellings of each item's reviewed Banglish (substitutions such as `sh`→`s`, `bh`→`v`, `kichhu`→`kichu`), preserving digits, formulae, option labels, and the answer line. Each item is then answered under all five spellings.

Table 4.7: Correctness stability across five Banglish spellings of the same 100 BEnQA items. A flip is an item whose correctness is not constant across spellings.

Model	Accuracy	Spelling range	Items flipping	Flip rate	Mean swing
Qwen2.5-3B	31.2%	28–34%	17/100	17%	0.17
Qwen3-4B	29.0%	28–31%	6/100	6%	0.06

Two findings emerge, and they point in complementary directions. First, the *magnitude* of Banglish accuracy is stable across independent spellings: per-spelling accuracy stays in a narrow 28–34% band for both models, so the script penalty is not an artifact of one particular romanizer's spelling choices. This directly addresses the naturalness limitation. Second, at the *item* level, robustness is not perfectly stable: 17% of items for Qwen2.5-3B (and 6% for Qwen3-4B) flip correctness when only the spelling changes, with the gold answer and question fixed. Banglish reliability is therefore both systematically lower and locally unstable under realistic spelling variation — the same user question can be answered or missed depending on how it happens to be romanized.

Chapter 5

Failure Analysis

The paired design lets us separate hard items from script-specific failures. This is the main advantage of keeping item ids fixed. Across the three Qwen models and 200 validation items, reviewed Banglish is wrong in 463 of 600 model-item slots. Of these Banglish misses, 185 are recoverable: either Bangla or English is correct on the same item. The remaining 278 are all-script hard.

Table 5.1: Recoverability source decomposition over 600 Qwen model-item slots

Category	Count
Reviewed Banglish wrong	463/600
Recoverable by Bangla or English	185/600
All-script hard	278/600
Bangla-only recovery	28/600
English-only recovery	81/600
Both alternate scripts recover	76/600
Banglish-only success	20/600

Figure 5.1 visualizes the decomposition: the lower bar splits the 463 Banglish misses into the recovery channels.

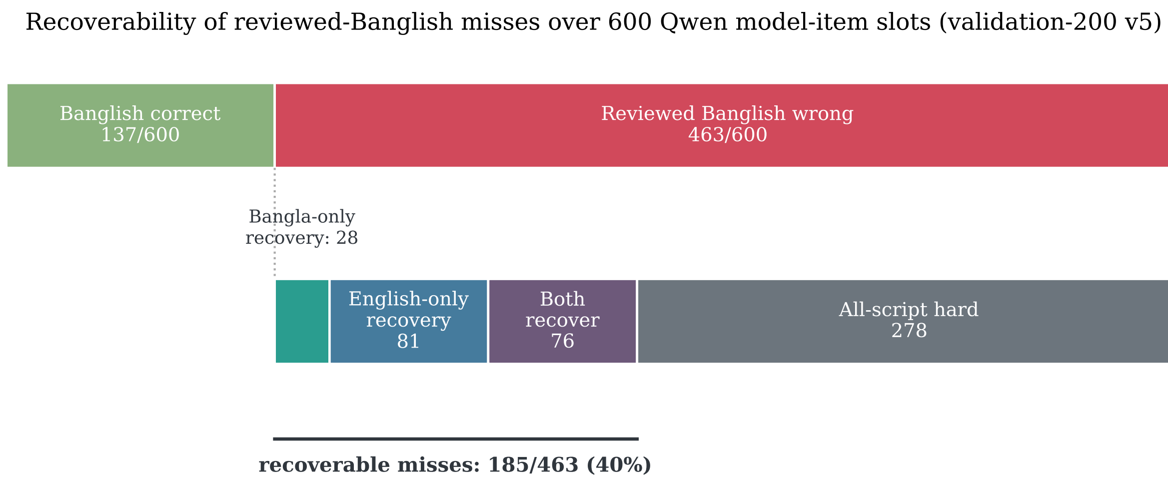

Figure 5.1: Recoverability decomposition of reviewed-Banglish misses over the 600 Qwen model-item slots of validation-200 v5. Of 463 Banglish misses, 185 (40%) are answerable by the same model when the same item is shown in Bangla or English.

This decomposition gives two important messages.

First, a large share of Banglish errors are not impossible items. In BEnQA, 55.2% of Banglish misses are recoverable by Bangla or English. These are the clearest qualitative examples for the thesis: the same answer becomes reachable when the script view changes.

Second, the result is not absolute. Banglish-only success exists. There are 20/600 model-item slots where reviewed Banglish is correct and both alternate views are wrong. We therefore avoid saying "Banglish is always worse." The claim is aggregate and paired: reviewed Banglish is systematically less reliable for the models and tasks studied.

Table 5.2: Example recoverable reviewed-Banglish misses from the BEnQA extension

Model	ID	Gold	Correct script views	Parsed answers	Banglish prompt fragment
Qwen2.5-3B	`...0100`	C	Bangla, English	BN=C; BG=D; EN=C	`akotin o mayosin...`
Qwen2.5-3B	`...0120`	D	Bangla, English	BN=D; BG=B; EN=D	`byakoteriyate...`
DeepSeek V4 Flash	`...0001`	C	Bangla, English	BN=C; BG=B; EN=C	`salokosongshleshoner...`
DeepSeek V4 Flash	`...0023`	A	Bangla	BN=A; BG=D; EN=D	`kshariy mutr...`

`BN`, `BG`, and `EN` denote parsed answers under Bangla, reviewed Banglish, and English prompts.

These examples should not be read as a separate test. They are explanatory evidence. They show the mechanism of the paired result: the same item and answer can become unreliable when the input is written in reviewed Banglish.

5.1 Error Taxonomy: Why Banglish Fails

The recoverability counts say that failures are not random difficulty, but they do not say what goes wrong. To answer the "but why does it fail?" question, we code every one of the 164 recoverable BEnQA reviewed-Banglish misses across the three Qwen models into four categories using documented deterministic rules (`scripts/analyze_banglish_error_taxonomy.py`). Because these are recoverable misses, the model demonstrably can answer the item in Bangla or English, so each category isolates a failure introduced by the romanization itself rather than item difficulty.

Table 5.3: Error taxonomy of the 164 recoverable BEnQA reviewed-Banglish misses across the three Qwen models.

Category	Count	Share
Named-entity / technical-term corruption	58	35.4%
Number / unit / formula misread	53	32.3%
Romanized-word ambiguity	51	31.1%
Option-format failure	2	1.2%
Total	164	100%

The categories are nearly balanced, which is itself informative: there is no single dominant failure mode, so a narrow fix will not close the gap.

Named-entity / technical-term corruption (35%) is the largest category. The decisive options are romanized scientific terms whose Bangla spelling encodes a precise concept. In `benqa_10th-Biology_0128` the options are `lyakotik esid`, `glukoj`, `oksalo asitik esid`, and `glisarik esid` (lactic acid, glucose, oxaloacetic acid, glyceric acid); Qwen2.5-3B answers correctly in Bangla and English but selects the wrong romanized term. The romanization is faithful, but the model fails to ground the Latin-script chemistry vocabulary back to the concept.

Number / unit / formula misread (32%) covers items that hinge on digits, units, or chemical formulae embedded in romanized text. In `benqa_10th-Biology_0215` the stem carries the glycolysis formula `C_{6}H_{12}O_{6}` → `C_{3}H_{4}O_{3}` alongside a romanized Roman-numeral option list; the model recovers the item in English but mis-selects under Banglish, where the mixed numeric-plus-romanized context is harder to parse.

Romanized-word ambiguity (31%) is the residual category where ordinary Bangla words become ambiguous in Latin script and shift the model toward a wrong but plausible option. `benqa_10th-Physics_0280` asks for copper's resistivity (`tamar rodhokotto`); the numeric options differ only at the second significant figure, and the model picks the wrong value under Banglish while answering correctly in English.

Option-format failure (1%) is rare here: only two recoverable misses produce an unparseable answer. The reviewed Banglish field almost always yields a clean A–D letter, so the gap is overwhelmingly about choosing the wrong option, not about violating the answer contract — which rules out a trivial parsing explanation. Full per-item codings are in `reports/banglish_error_taxonomy.md`.

Chapter 6

Frontier and Scale Boundary

6.1 Within-Family Scaling: Does the Gap Grow with Capability?

Before turning to hosted frontier models, we ask a sharper question inside one model family: as a Qwen model gets larger, does the Banglish gap shrink (capability helps) or grow (capability is spent on the easier scripts)? We run the frozen validation-200 v5 triad on four additional Qwen sizes — Qwen2.5-0.5B and 1.5B, and Qwen3-0.6B and 1.7B (no-thinking) — and combine them with the three thesis-facing models on the same frozen slice.

Table 6.1: Within-family Qwen scaling on validation-200 v5. Gap is reviewed Banglish minus Bangla, with paired bootstrap 95% CI and McNemar exact p.

Model	Family	Params (B)	Bangla	Banglish	Gap (pts)	McNemar p
Qwen2.5-0.5B	Qwen2.5	0.49	40/200	46/200	+3.0, CI [-1.5, +7.5]	0.263
Qwen2.5-1.5B	Qwen2.5	1.54	46/200	40/200	-3.0, CI [-9.0, +3.0]	0.418
Qwen2.5-3B	Qwen2.5	3.09	54/200	41/200	-6.5, CI [-13.0, 0.0]	0.066
Qwen2.5-7B 8-bit	Qwen2.5	7.62	65/200	47/200	-9.0, CI [-16.5, -1.5]	0.022
Qwen3-0.6B	Qwen3	0.60	35/200	29/200	-3.0, CI [-8.0, +2.0]	0.345
Qwen3-1.7B	Qwen3	1.72	33/200	35/200	+1.0, CI [-5.5, +7.5]	0.878
Qwen3-4B	Qwen3	4.02	80/200	49/200	-15.5, CI [-22.0, -9.0]	<0.0001

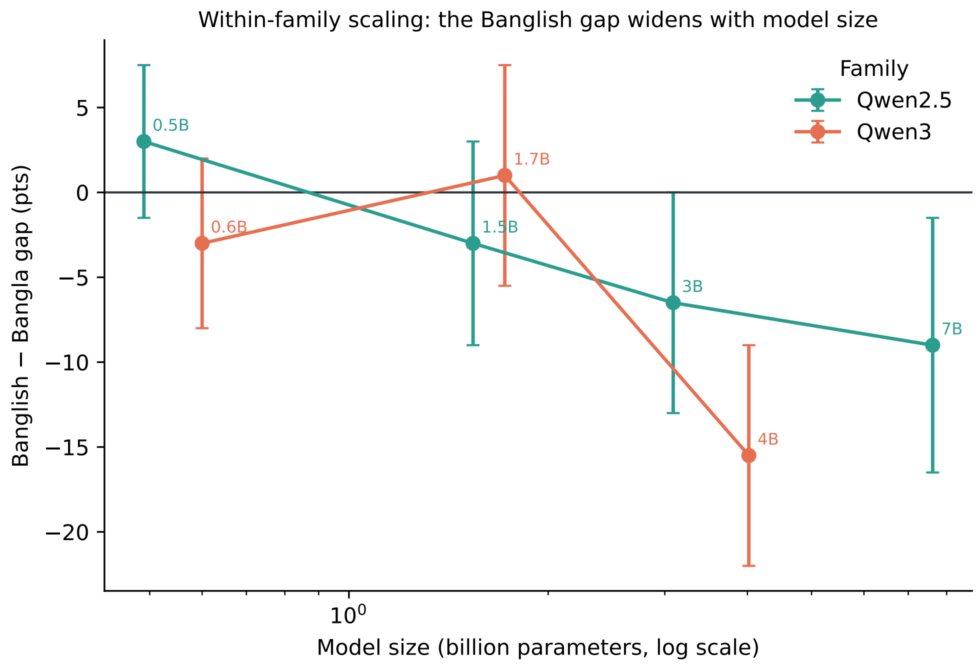

Figure 6.1: Banglish minus Bangla gap versus model size within the Qwen2.5 and Qwen3 families on validation-200 v5, with paired bootstrap 95% confidence intervals.

The Qwen2.5 family shows a clean monotonic trend: the gap moves from $+3.0$ points at 0.5B (where the model is too weak for the contrast to mean much) through -3.0, -6.5, and -9.0 points at 1.5B, 3B, and 7B, and the McNemar p-value falls correspondingly from 0.26 to 0.02. The gap does not close with capability; it widens. The Qwen3 family is noisier at the bottom — the 1.7B no-thinking point is weak and near zero — but its largest model has by far the largest gap (-15.5 points, $p < 0.0001$). The reading is consistent with the thesis mechanism: added capability is preferentially spent on the higher-resource Bangla and English views, so the romanized-script deficit persists or grows rather than being absorbed. This makes the script gap a property that scaling alone, within these compact families, does not fix.

6.2 Frontier API Boundary

The frontier/API panel tests whether the gap disappears for stronger or hosted models. The answer is not a simple yes or no. Stronger models can reduce the gap, but the reduction is uneven and model-dependent.

Table 6.2: Validation-200 v5 frontier/API panel. Difference columns report reviewed Banglish minus the comparison script.

Model	Score	Bangla	Reviewed Banglish	English	Banglish–Bangla	Banglish–English
Gemini 3.5 Flash	Strict	163/200	136/200	144/200	-13.5 pts	-4.0 pts
Gemini 3.5 Flash	Secondary	170/200	161/200	165/200	-4.5 pts	-2.0 pts
GPT-5.5 low	Strict	172/200	169/200	154/200	-1.5 pts	+7.5 pts
GPT-5.5 low	Secondary	173/200	174/200	168/200	+0.5 pts	+3.0 pts
Claude Sonnet 4.6	Strict	162/200	130/200	153/200	-16.0 pts	-11.5 pts
Claude Sonnet 4.6	Secondary	167/200	133/200	166/200	-17.0 pts	-16.5 pts
DeepSeek V4 Flash	Strict	143/200	82/200	132/200	-30.5 pts	-25.0 pts
DeepSeek V4 Flash	Secondary	152/200	96/200	148/200	-28.0 pts	-26.0 pts
Groq Llama 3.3 70B	Strict	90/200	48/200	102/200	-21.0 pts	-27.0 pts
Groq Llama 3.3 70B	Secondary	92/200	56/200	111/200	-18.0 pts	-27.5 pts

GPT-5.5 low is the strongest boundary case. Under strict scoring it nearly closes the Banglish-vs-Bangla gap: 172/200 Bangla versus 169/200 Banglish. Under secondary scoring, Banglish is slightly ahead of Bangla. This is an important limitation on the main claim. With enough capability and answer recovery, the reviewed-Banglish gap can nearly disappear on the validation-200 set.

The McNemar exact tests make the panel contrast precise. GPT-5.5's strict Bangla-versus-Banglish discordance is 6 losses against 3 gains ($p = 0.51$): statistically indistinguishable from no gap. Every other API row is significant after Holm correction across the panel ($p < 0.0001$ for Gemini, Claude, DeepSeek, and Groq under strict scoring), with conditional odds ratios from 3.9 (Groq) to 9.1 (DeepSeek). The full per-panel tables are in `reports/mcnemar_script_gaps.md`.

But that is not the whole frontier story. Gemini 3.5 Flash reduces the gap but retains a strict deficit. Claude Sonnet 4.6 has high absolute accuracy but still scores 32 items lower on reviewed Banglish than on Bangla under strict scoring. It also produces more long and multiline answers under an answer-only prompt, which creates format-compliance and cost issues. DeepSeek V4 Flash and Groq-hosted Llama 3.3 70B retain large Banglish deficits.

The correct conclusion is therefore not "frontier models solve Banglish." It is more precise: stronger models can reduce the semantic gap, but script robustness remains model-dependent, parser-dependent, and cost-dependent.

6.3 BEnQA Scale Extension

The validation-200 reviewed core is deliberately small and carefully audited. That is a strength for controlled analysis, but it also raises an obvious scale question: does the pattern survive beyond the selected 200 items? We answer that with a larger BEnQA-only human-reviewed extension.

The extension construction starts from 4,939 eligible BEnQA source rows after excluding the frozen gold-core BEnQA items. We select 1,000 rows with deterministic subject-balanced sampling. The human review audit accepts 618 rows unchanged, edits 356 rows, and rejects 26 rows. The 974 accepted or edited rows form the BEnQA human-reviewed gold extension used for large-scale evaluation.

This extension does not replace validation-200 v5 because it is BEnQA-only and therefore covers science multiple-choice questions rather than the mixed BEnQA-plus-BanglaMATH task design. Its role is scale: combined with the validation core, the thesis now has 1,174 usable human-reviewed paired items, while preserving the 200-item core as the mixed-task primary benchmark.

Table 6.3: BEnQA human-reviewed gold 974 scale results. The final column reports the Holm-adjusted McNemar exact p-value for the Bangla-versus-Banglish contrast.

Model	Bangla	Reviewed Banglish	English	Banglish–Bangla	Banglish–English	McNemar p
Qwen2.5-3B	323/974	285/974	490/974	-3.90 pts, CI [-7.19, -0.51]	-21.05 pts, CI [-24.64, -17.35]	0.024
Gemini 3.5 Flash	743/974	633/974	680/974	-11.29 pts, CI [-13.66, -8.93]	-4.83 pts, CI [-7.39, -2.05]	<0.0001
GPT-5.5 none	820/974	699/974	825/974	-12.42 pts, CI [-15.09, -9.75]	-12.94 pts, CI [-15.91, -9.86]	<0.0001
Claude Sonnet 4.6	764/974	524/974	771/974	-24.64 pts, CI [-27.82, -21.25]	-25.36 pts, CI [-28.64, -21.87]	<0.0001
DeepSeek V4 Flash	756/974	438/974	791/974	-32.65 pts, CI [-36.04, -29.26]	-36.24 pts, CI [-39.73, -32.96]	<0.0001
Groq Llama 3.3 70B	547/974	333/974	622/974	-21.97 pts, CI [-25.67, -18.17]	-29.67 pts, CI [-33.26, -26.08]	<0.0001

All six 974-row extension runs reproduce the same core pattern: **reviewed Banglish is the weakest script view for every model**, with every Bangla-versus-Banglish contrast significant under the Holm-corrected McNemar exact test. The compact open model and the Groq, DeepSeek, and GPT-5.5 rows additionally show the full English > Bangla > reviewed Banglish ordering; Gemini is the one model whose Bangla accuracy exceeds its English accuracy, which does not change the Banglish conclusion.

The most consequential row is GPT-5.5. On validation-200 it nearly closes the gap; on the 974-row extension, run with no reasoning effort, it loses 12.42 points on reviewed Banglish relative to Bangla ($p < 0.0001$). The near-closure observed on the 200-item core therefore does not generalize to the larger slice under a low-compute configuration. Script robustness at the frontier is not a solved property; it depends on the model, the reasoning budget, and the evaluation scale.

The models behave differently in absolute accuracy. DeepSeek V4 Flash is far stronger than Qwen2.5-3B on the extension. But the Banglish deficit remains, and for DeepSeek it is much larger. This is one of the strongest results in the thesis. It shows that higher absolute accuracy does not guarantee Banglish robustness, and that the extension result is not a Qwen-local artifact. One operational caveat: the Gemini extension run has elevated parsed-empty counts (176–263 rows per variant), so its absolute scores are less clean than its paired contrast.

The 974-row runs also provide qualitative examples. Qwen2.5-3B has 346 recoverable reviewed-Banglish misses, Groq-hosted Llama 3.3 70B has 406, DeepSeek V4 Flash has 424, Claude Sonnet 4.6 has 330, GPT-5.5 has 199, and Gemini 3.5 Flash has 166. These are cases where reviewed Banglish is wrong but Bangla or English is correct on the same item. They are useful for error analysis and defense slides, but they are not a separate statistical test.

6.3.1 Where the extension gap concentrates: subject breakdown

The 974-row extension is large enough to split by macro subject (Biology 229, Physics 226, Math 223, Chemistry 221, Science 75 items). Table 6.4 reports the Banglish-minus-Bangla gap per subject.

Table 6.4: BEnQA human-gold 974 extension: reviewed Banglish minus Bangla gap in points by macro subject. Stars mark within-subject McNemar exact $p < 0.05$.

Model	Biology	Chemistry	Physics	Math	Science
Qwen2.5-3B	-3.1	-5.4	-6.6	-2.7	+2.7
Gemini 3.5 Flash	-13.1*	-14.5*	-10.2*	-10.8*	-1.3
GPT-5.5 none	-17.9*	-14.9*	-13.3*	-3.6	-12.0*
Claude Sonnet 4.6	-33.2*	-25.8*	-30.1*	-6.3*	-33.3*
DeepSeek V4 Flash	-38.0*	-39.4*	-37.2*	-13.5*	-40.0*
Groq Llama 3.3 70B	-26.2*	-24.0*	-24.3*	-9.9*	-32.0*

Full per-subject accuracy and discordant-pair tables:
`reports/benqa_human_gold_974_subject_breakdown.md`.

The breakdown reveals a consistent structure. For every frontier model, the Banglish gap is smallest on Math and largest on the text-heavy science subjects, with Biology and Science often worst. GPT-5.5's Math gap is statistically indistinguishable from zero (-3.6 points, $p = 0.28$) while its Biology gap is -17.9 points ($p < 0.0001$). DeepSeek loses around 38–40 points on the science subjects but 13.5 on Math. The pattern matches the mechanism the thesis proposes: romanization damages lexical grounding of Bangla terminology, and items whose content survives in symbolic or numeric form are less exposed. The script gap is therefore not uniform difficulty inflation; it concentrates where the question depends on reading romanized Bangla words.

6.4 Natural Code-Mixed External Layer

The controlled benchmark uses curriculum-style QA and math. That gives strong paired evidence, but it does not fully represent natural social media Banglish. We therefore add a separate ecological-validity layer using BnSentMix, a Bengali-English code-mixed sentiment dataset [12].

We build a balanced 200-row four-way sentiment slice with 50 positive, 50 negative, 50 neutral, and 50 mixed items. The task is not paired by script, so it cannot estimate a Bangla-vs-Banglish script penalty. It answers a different question: do compact open models handle naturally occurring Bengali-English mixed text reliably?

Table 6.5: BnSentMix natural code-mixed sentiment results

Model	Accuracy	Macro-F1	Valid outputs
Qwen2.5-3B	89/200	0.431	200/200
Qwen2.5-7B 8-bit	98/200	0.479	200/200
Qwen3-4B	99/200	0.486	200/200

The models are above the 25% majority baseline, but absolute accuracy remains modest. The error-overlap analysis shows substantial complementarity: the best single model is 99/200, while a diagnostic any-model oracle reaches 154/200. This oracle is not deployable because it chooses the correct model after seeing the gold label. Its value is diagnostic: model errors on natural code-mixed text are not uniform.

This layer supports the broader motivation that Bengali-English mixed text is difficult and user-relevant. It does not replace the controlled paired script-gap result.

Chapter 7

Mitigation Attempts

The mitigation story should be honest. The thesis does not yet have a general solution, and that negative result is still useful: it prevents the evaluation problem from being hidden behind a fragile prompt wrapper.

7.1 Self-normalization

Same-model self-normalization asks the model to rewrite Banglish into Bangla and then answer. It helps one model and hurts another.

Table 7.1: Self-normalization on validation-200 v3

Model	Clean Banglish baseline	Self-normalized	Delta
Qwen2.5-3B	38/200	51/200	+6.5 pts
Qwen3-4B	46/200	21/200	-12.5 pts

The Qwen2.5-3B gain is real but brittle. It includes 27 gains and 14 losses, and the rewrite audit shows option, digit, and formula preservation errors. The Qwen3-4B result is more severe: accuracy collapses even though surface preservation counters look better. This means the procedure changes model decision behavior, not just text quality.

The conclusion is simple: self-normalization is not a general Banglish solution.

7.1.1 Generated alternate-script views

A stronger mitigation idea is to generate alternate script views and route only when views agree. This is attractive because the cross-script oracle shows that many Banglish misses are recoverable under Bangla or English. But the deployable version cannot use gold benchmark views. It must generate those views reliably.

The generated-view audits show why this is hard.

Raw deterministic Bangla transliterators corrupt MCQ option labels. Protected variants reduce some failures, but earlier versions fail tightened formula-expression gates. A formulaish-token protected-v3 wrapper repairs the hard preservation gate on the small dev audit, but answer gains remain small and uncertain. A guarded generated-English repair passes preservation gates, but 15/36 rows fall back to the source Banglish text. Agreement routing is too sparse: it misses most generated-view oracle recoveries.

The correct claim is therefore negative:

Generated-view routing is promising, but current cheap generated views are not reliable enough for a held-out mitigation claim.

This matters because it prevents overclaiming. The thesis diagnoses a real orthographic robustness problem and shows that naive mitigation is brittle.

7.2 Training-Side Mitigation: LoRA Adaptation

The previous mitigations act at inference time. A stronger question is whether a small amount of romanized supervision can close the gap from the training side. We train two LoRA adapters on Qwen2.5-3B-Instruct (rank 16, $\alpha = 32$, dropout 0.05, on all attention and MLP projections; 0.96% of parameters; two epochs, fp16). The training data is a BEnQA pool of 3,739 items whose source rows are disjoint from both the frozen validation-200 v5 core and the 1,000-row extension, with the disjointness asserted in code before any file is written. Arm A trains on Banglish-only completions; arm B trains on a 1:1:1 Bangla/Banglish/English mix, which tests whether balanced supervision avoids forgetting the other scripts. A held-out 200-item dev split (never trained) verifies that learning occurred.

On the dev split the adapters do learn: arm A raises Banglish from 63/200 to 75/200 ($+6.0$ points) and arm B raises it to 70/200, with neither Bangla nor English regressing (arm B improves all three views). The dev Banglish gains are positive but not individually significant (arm A McNemar $p = 0.19$), so the sanity check passes without overclaiming.

The decisive test is the frozen validation-200 v5 triad, rerun with each merged adapter under the same prompt manifest and parser.

Table 7.2: LoRA adaptation on the frozen validation-200 v5 triad. The gap is reviewed Banglish minus Bangla. Deltas versus base are paired; none is statistically significant.

Condition	Bangla	Banglish	English	Banglish – Bangla gap
Base (no adapter)	54/200	41/200	71/200	-6.5 pts
Arm A (Banglish-only)	57/200	46/200	69/200	-5.5 pts
Arm B (mixed 1:1:1)	55/200	47/200	76/200	-4.0 pts

Both adapters move the gap modestly toward zero — from -6.5 points at base to -5.5 (arm A) and -4.0 (arm B) — and, importantly, neither Bangla nor English regresses, so there is no catastrophic forgetting. But the movement is small and not statistically significant: the paired Banglish gain over base is $+2.5$ points for arm A (CI $[-3.5, +8.5]$, $p = 0.50$) and $+3.0$ points for arm B (CI $[-2.5, +8.5]$, $p = 0.36$). Moreover, arm B lifts Banglish, Bangla, and English together, which is the signature of general task adaptation rather than script-specific mitigation. Full numbers are in `reports/lora_mitigation_results.md`.

The honest reading follows the success gates set before the experiment. This is not a positive mitigation result: the Banglish gap interval does not exclude zero, and the dev-split gains do not translate into a significant frozen-slice reduction. It is a partial, adaptation-level effect at most. As a negative mitigation result it is informative: light, eval-disjoint LoRA supervision — the natural first training-side fix — narrows the Banglish gap only slightly and not reliably, which strengthens rather than weakens the thesis. The orthographic weakness is deeper than a small romanized-supervision patch can close, and a genuine fix likely requires more romanized pretraining exposure or architecture-level changes, not a light adapter at fine-tuning time.

Chapter 8

Discussion, Limitations, and Artifacts

8.1 Discussion

8.1.1 What the result means

The result should be read with the right scope. It is not that Bangla is weak. It is not that Banglish is always worse. It is not that stronger models never solve the problem.

The result is that script choice changes reliability under controlled conditions. A model can answer an item in Bangla or English and fail on reviewed Banglish. This happens often enough to affect aggregate accuracy, paired gaps, and qualitative examples. It survives review, tokenization controls, compact model scaling, hosted-model comparison, and evaluation on the larger BEnQA extension.

The practical lesson is similar to many robustness studies: the visible system may look multilingual, but the interface can still contain a hidden access failure. Here the hidden variable is not the topic or the answer; it is the orthography through which the user reaches the model.

8.1.2 Why GPT-5.5 matters

GPT-5.5 is the strongest boundary case. Under secondary scoring, it nearly eliminates the reviewed-Banglish gap on the validation-200 set. This is important and should not be hidden. It tells us that high-capability models can recover much of the semantic answer.

But GPT-5.5 does not make the broader problem disappear. First, strict scoring still reveals answer-normalization and protocol issues. Second, other strong or hosted models do not behave like GPT-5.5. Claude still shows a Banglish deficit, and DeepSeek and Groq retain large deficits. Third, even when the semantic answer is recoverable, cost and format compliance can change with script. Fourth, and most importantly, the near-closure does not persist at scale: on the 974-row BEnQA extension, run with no reasoning effort, GPT-5.5 loses 12.4 points on reviewed Banglish relative to Bangla with a McNemar exact $p < 0.0001$. For deployment, the question is not only "can any model answer?" but "which models answer reliably, cheaply, and in the required format — and does that hold beyond a small audited core?"

8.1.3 Why the human-reviewed BEnQA extension matters

The 974-row human-reviewed BEnQA extension is important because it separates model capability from script robustness at a larger scale. DeepSeek V4 Flash is much stronger than Qwen2.5-3B on the BEnQA extension. Yet its reviewed-Banglish deficit is much larger. The five non-Qwen scale rows — Gemini, GPT-5.5, Claude, DeepSeek, and Groq-hosted Llama — all reproduce a significant reviewed-Banglish deficit. This makes the thesis stronger than a small-model failure story. A model can be high-accuracy in Bangla and English while still being fragile in Banglish.

8.1.4 Practical implications

For benchmark builders, script variants should be first-class conditions, not informal noise. If a benchmark only measures native script or English, it can miss a real access gap for users who write in Latin-script Banglish.

For model developers, normalization should be evaluated end to end. A transliterator that preserves words may still corrupt options, digits, formulas, or decision behavior. Preservation gates are necessary before routing or generated-view mitigation claims.

For deployment, a model that is strong in Bangla is not automatically robust to Banglish. Applications that serve Bangla users should log, test, and report performance separately for native Bengali script and Latin-script Banglish.

8.2 Considerations of Professional Ethics, Responsibilities and Norms of Engineering Practice

This thesis was conducted as a measurement study, so the main ethical responsibility was to make the evaluation fair, reproducible, and honest about its limits. The study compares the same underlying questions across Bangla, reviewed Banglish, and English views. The benchmark construction therefore avoided changing the question, answer, or scoring rule when changing the script. This preserves the central fairness principle of the work: a script gap should reflect model behavior under different orthographic access paths, not hidden changes in item difficulty.

Research ethics were maintained by using documented datasets, preserving source attribution, and reporting both successful and unsuccessful outcomes. Rows that raised source-quality concerns were not silently removed from the main denominator after seeing results; instead, the all-200 denominator was kept as primary and the strict-197 view was reported as a sensitivity check. Similarly, the thesis reports that GPT-5.5 nearly closes the validation-200 Banglish gap under secondary scoring, even though the broader scale result remains unfavorable to Banglish. This is important because an ethical evaluation should not hide evidence that complicates the thesis claim.

The work did not collect private user conversations or personally identifiable information. Human review was used to improve benchmark quality, especially the reviewed Banglish field, but the review target was linguistic and task preservation rather than collection of sensitive personal data. The reviewed artifacts, analysis scripts, tables, and reports are listed in the reproducibility section so that future readers can inspect how the claims were produced.

The engineering practices followed conventional experimental norms for machine learning evaluation. The experiment was designed as a paired comparison, where each item is evaluated under multiple script views with the same gold answer. This reduces confounding and makes statistical tests such as paired bootstrap intervals and exact McNemar tests appropriate. The evaluation protocol froze the validation set before reporting final results, used consistent prompts and parsers across script conditions, separated strict scoring from secondary recoverability scoring, and preserved logs and output artifacts for audit.

Coding and testing responsibilities were also treated as part of the engineering process. Scripts were written to build datasets, run evaluations, import API responses, compute tables, check artifact references, and validate thesis figures and tables. Outputs were cross-checked through summary reports, reproducibility manifests, local artifact checks, and targeted sensitivity analyses. These checks reduce the risk of accidental data leakage, mismatched denominators, broken file references, or unsupported claims. The resulting thesis is therefore not only a set of model scores but an engineered evaluation pipeline with traceable inputs, procedures, and outputs.

8.3 Reflection on Project Management, Team Dynamics, and Economic Decision-Making

The project was managed as a sequence of research milestones. The first milestone was to define the research question: whether Latin-script Banglish changes LLM reliability compared with native-script Bangla and English. The second milestone was benchmark construction, including source-task selection, script-variant generation, human review, and denominator policy. The third milestone was model evaluation and robustness analysis across compact open models. Later milestones extended the work through hosted-model comparison, the larger human-reviewed BEnQA extension, mitigation attempts, artifact checking, and thesis writing.

Because this was a single-student thesis, team dynamics occurred mainly through supervisor guidance, thesis meetings, feedback cycles, and follow-up revisions rather than through division of labor among multiple students. Meetings and discussions helped refine the scope of the work, identify weak claims, decide which experiments were necessary, and prioritize results that were suitable for an undergraduate thesis timeline. After each major result, the work was followed up by checking whether the claim needed additional review, a stronger statistical test, a clearer limitation, or a better artifact trail.

The project also required coordination between research tasks that depended on each other. Dataset review had to be completed before final model evaluation could be treated as frozen. API experiments had to use the same prompt manifest and parser as the local experiments. Thesis tables and figures had to be rebuilt only after the analysis outputs were stable. This dependency structure made project management important: running experiments too early could waste compute, while delaying analysis too long could leave insufficient time for writing and verification.

Economic decision-making was relevant because LLM evaluation has real computational and financial costs. The thesis therefore used compact Qwen models as the main reproducible baseline, since they could be evaluated with available compute while still producing meaningful evidence. More expensive hosted-model calls were reserved for targeted validation rather than unlimited exploration. Smoke tests, prompt-budget estimates, capped runs, and manifest checks were used before larger API jobs to reduce unnecessary spending and avoid repeating failed runs. This allowed the thesis to balance research value against cost, time, and available hardware.

The financial aspect of the study was therefore not only a matter of direct API fees. It also included GPU time, storage for artifacts, repeated evaluation time, and the opportunity cost of running experiments that might not answer the research question. The chosen strategy was to spend resources where they strengthened the core claim: paired script evaluation, human review, statistical testing, frontier-model boundary checks, and the BEnQA scale extension.

8.4 Reflection on Self-directed and Life-long Learning to Navigate Technological Change

This thesis required substantial self-directed learning because the research area changed quickly during the work. The study involved multilingual evaluation, Bangla and Banglish text processing, LLM prompting, answer parsing, statistical significance testing, API-based model evaluation, and reproducibility engineering. Many of these components required learning beyond regular coursework and then applying that learning to a coherent experimental pipeline.

The self-directed part of the thesis is visible in the construction of the benchmark and analysis workflow. The work required learning how to design paired evaluations, identify confounds in script comparison, audit romanized text quality, interpret bootstrap intervals and McNemar tests, and separate strict answer compliance from semantic recoverability. It also required learning how to manage model outputs from both local inference and hosted APIs, how to preserve artifacts for future inspection, and how to revise claims when new evidence changed the interpretation.

The topic itself demonstrates the need for life-long learning in engineering practice. LLM capabilities, model families, tokenizers, API behavior, and multilingual benchmarks evolve rapidly. A result that holds for one generation of models may weaken, strengthen, or change form in another. For example, GPT-5.5 behaves differently from several other hosted models on the validation-200 core, but the larger BEnQA extension still reveals a Banglish deficit. This means engineers cannot rely on a fixed assumption such as “newer models solve multilingual input.” They must continue measuring real user-facing conditions as systems change.

The thesis also helped develop habits that support future learning: reading current literature, building reproducible experiments, validating assumptions with data, documenting artifacts, and treating negative or unresolved results as useful evidence. The mitigation experiments are an example. Generated-view routing and light LoRA adaptation did not solve the problem, but they clarified what kind of preservation and robustness requirements a future solution must satisfy.

Beyond this study, the same learning process can be applied to other low-resource or mixed-script settings. Engineers working with language technologies need to update their methods as datasets, user behavior, and model capabilities change. This thesis therefore served not only as a study of Banglish robustness but also as an exercise in adapting to technological change through self-directed investigation, careful measurement, and continuous revision of engineering assumptions.

8.5 Limitations

This thesis has several limitations. These limitations are important because they define what the results should and should not be used to claim.

Controlled Banglish is not fully natural Banglish. The reviewed Banglish variant is designed for paired evaluation. It is cleaner and more task-like than many user messages. Real Banglish can be shorter, noisier, more code-mixed, and more spelling-variable. The spelling-variation experiment (Table 4.7) partly addresses this: the script penalty is stable in magnitude across five independent phonetic respellings of each item, so it is not an artifact of one romanizer, although a meaningful minority of items flip correctness under spelling change. BanglaTLit and BnSentMix further motivate this, but they do not replace a future naturally paired Bangla/Banglish QA dataset.

The validation set is small by benchmark standards. Validation-200 is a reviewed mixed-task core, not a massive benchmark. Its strength is the paired design, review, and audits. The 974-row BEnQA extension addresses scale for one task family and uses only rows accepted or edited during human review, but it does not add a similarly large BanglaMATH slice.

The extension is BEnQA-only. The 974-row scale result covers science MCQs. It does not scale the BanglaMATH part of the benchmark. BanglaMATH remains a smaller stress-test slice for arithmetic and answer normalization.

Model coverage is broad but not exhaustive. The thesis covers compact Qwen models and five hosted API rows, but it does not cover every family. It should not claim universal behavior across all LLMs. The correct claim is that script robustness is model-dependent and must be measured.

Secondary scoring is not a replacement for strict scoring. Secondary scoring helps separate semantic recoverability from format compliance. But in deployed systems, strict format compliance often matters. Both views should be reported.

Mitigation remains unresolved. The generated-view experiments are dev-only diagnostics that show preservation and routing problems, not a deployable fix. The training-side LoRA experiment is a single 3B model with one rank, two epochs, and a BEnQA-only training pool; it bounds light adaptation but does not rule out that larger adapters, more romanized data, or additional model families could help.

8.6 Reproducibility and Artifacts

The core artifacts are:

- Main validation-200 v5 report: `reports/main_results_validation200_v5.md`
- Main validation table: `results/tables/main_script_gap_validation200_v5.csv`
- Frontier API panel: `reports/frontier_api_panel_validation200_v5.md`
- BEnQA human-reviewed extension freeze: `reports/benqa_extended_1000_v1_human_review_freeze.md`
- BEnQA human-reviewed 974 scale summary: `reports/benqa_human_gold_974_scale_summary.md`
- Qwen2.5-3B human-reviewed 974 result: `reports/qwen25_3b_benqa_human_gold_974.md`
- Groq Llama 3.3 70B human-reviewed 974 result: `reports/groq_llama33_70b_benqa_human_gold_974.md`
- DeepSeek V4 Flash human-reviewed 974 result: `reports/deepseek_v4_flash_benqa_human_gold_974.md`
- BEnQA extension strategy: `reports/benqa_extension_publication_strategy.md`
- BnSentMix external layer: `reports/bnsentmix_external_validation_results.md`
- Generated-view diagnostics: `reports/generated_view_diagnostics_summary.md`
- Dataset card: `reports/dataset_card_validation200.md`

The current status dashboard reports 71 rows, 0 blocked, and 0 failing:

- `reports/current_research_status_dashboard.md`

The local artifact reference checker reports 0 unexpected missing references, and the reproducibility manifest tracks 1116 artifacts:

- `reports/local_artifact_reference_check.md`
- `reports/reproducibility_artifact_manifest.md`

Chapter 9

Conclusion

Script choice matters for Bangla LLM evaluation. On a controlled Bangla/Banglish/English validation set, reviewed Latin-script Banglish is less reliable than native Bengali script and English for compact Qwen models. The gap survives targeted review, the strict-197 sensitivity check, tokenization audits, recoverability analysis, and Holm-corrected McNemar exact tests. Frontier models reduce the gap unevenly: GPT-5.5 nearly closes it on the 200-item core, but Claude, DeepSeek, and Groq preserve clear deficits under the same protocol. A 1,000-row human review audit adds a 974-row accepted/edited BEnQA gold extension, and full runs with Qwen2.5-3B, Gemini 3.5 Flash, GPT-5.5, Claude Sonnet 4.6, DeepSeek V4 Flash, and Groq-hosted Llama 3.3 70B all reproduce a statistically significant reviewed-Banglish deficit outside the 200-item reviewed core — including GPT-5.5, whose near-closure on the core does not persist at scale.

Mitigation remains unresolved on both sides of the model. Inference-time normalization and generated-view routing are brittle, and a training-side LoRA adapter with eval-disjoint romanized supervision narrows the gap only slightly and not significantly, without regressing the other scripts. The orthographic weakness is therefore deeper than a light adaptation patch can close.

The broader lesson is practical. Banglish should not be treated as informal noise that models will automatically handle. For Bangla users, Latin-script input is a real access path. Robust evaluation must measure it directly, and robust mitigation must preserve task structure, answer format, and script equivalence rather than assuming that larger models or simple normalization will solve the problem.

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